

Siddhikesh Arvind Gavit

AI/ML Engineer | Generative AI | Agentic AI | LLM Applications | Gandhinagar, India

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SUMMARY

Final-year CSE undergraduate building LLM applications, agentic AI systems and retrieval-augmented generation (RAG) pipelines in Python. Shipped a LangGraph + MCP multi-agent assistant with human-in-the-loop approvals, a hybrid-search RAG app with cited answers, and an INT8-quantized model running on a microcontroller. Targeting AI Engineer, GenAI and ML Engineer roles.

EDUCATION

Indian Institute of Information Technology (IIIT) Vadodara

Aug 2023 – May 2027

B.Tech in Computer Science and Engineering; CPI: 6.33/10

Gandhinagar, Gujarat

Coursework: Machine Learning, Deep Learning, Artificial Intelligence, Data Structures & Algorithms, DBMS, Operating Systems, OOP

Hon. Sharad Pawar International School (CBSE): Class XII 80.00% (2022), Class X 84.80% (2020)

Nashik, Maharashtra

TECHNICAL SKILLS

Generative AI, LLMs & NLP: LLM APIs (OpenAI, Anthropic Claude, Google Gemini, Groq), Ollama, Hugging Face, Prompt Engineering, Tool/Function Calling, Structured Outputs, Model Context Protocol (MCP), Guardrails, Prompt-Injection Defense

Agents & RAG: LangGraph, LangChain, CrewAI, LlamaIndex, Multi-Agent Systems, Human-in-the-Loop, RAG, Hybrid Search (BM25 + Semantic), Reranking, Embeddings, Vector Databases (ChromaDB, Pinecone), LLM Evaluation (RAGAS)

Machine Learning: scikit-learn, TensorFlow, Keras, XGBoost, LSTM, GRU, TCN, Transformers, TensorFlow Lite Micro, INT8 Quantization, SHAP, LIME, Pandas, NumPy, Matplotlib, Feature Engineering, EDA, Statistical Testing

Backend & MLOps: FastAPI, Flask, REST APIs, Pydantic, Docker, GitHub Actions (CI/CD), pytest, Streamlit, Git, Postman

Languages & Tools: Python, SQL, Java, C++; PostgreSQL, MySQL, MongoDB, Supabase; n8n, Zapier, Make.com

EXPERIENCE

Research Intern, Edge AI and On-Chip Sleep-Quality Prediction

Jun 2026 – Aug 2026

IIIT Vadodara | [GitHub](#)

Gandhinagar, Gujarat

- Built a TinyML deployment pipeline: redesigned the model as a 1,425-parameter 1D-CNN, applied full INT8 post-training quantization (TensorFlow Lite Micro) and exported the 7 KB model as a C header, losing only 0.01 AUROC (0.813 to 0.803).
- Wrote C++ inference firmware for the Arduino Nano 33 BLE (ARM Cortex-M4) with on-device normalization, int8 (de)quantization and a 12 KB tensor arena: about 8 ms per prediction using 25 KB of 256 KB RAM.
- Replicated Lee et al. (2025) on a simulated 82-subject wearable cohort: benchmarked 7 models (LSTM, GRU, TCN, Transformer, XGBoost, Random Forest, ARIMA) on 7-day HRV windows; an LSTM + XGBoost ensemble led (AUROC 0.870).
- Kept evaluation leakage-free with subject-wise splits and a label-shuffle audit (AUROC 0.50), tuned decision thresholds with Youden's J, and explained predictions with SHAP and LIME (Python, TensorFlow, scikit-learn).

PROJECTS

Synapse: Personal Agentic AI Assistant | *Python, LangGraph, CrewAI, MCP, FastAPI, React, Docker*

[GitHub](#)

- Built a multi-agent assistant that turns natural-language goals into verified actions: a checkpointed LangGraph state machine runs plan, execute and verify with bounded replanning across 8 specialist agents and 5 CrewAI crews.
- Exposed 38 tools over 7 MCP servers (Obsidian, Gmail, Calendar, Notion, n8n, web search) with JSON Schema validation, timeouts and risk classes; send, delete and overwrite actions wait for human approval via LangGraph interrupts.
- Defended against prompt injection with plan-then-execute and taint tracking; traced tokens and latency per run (p50/p95), fell back from Groq GPT-OSS 120B to local Ollama on failure, and gated CI on 70+ pytest tests and 16 behaviour evals.

notes-gpt: Citation-Grounded PDF RAG Assistant | *Python, ChromaDB, BM25, Groq, RAGAS, Streamlit* [GitHub](#) · [Live Demo](#)

- Implemented RAG without a framework: PyMuPDF page-level parsing, overlapping chunks with source and page metadata, and answers that cite [file p.N] for every claim or say "I don't know" when the context is missing.
- Built hybrid retrieval merging BM25 and semantic search (bge-small-en-v1.5 in ChromaDB) with bge-reranker-base cross-encoder reranking; answers stream from GPT-OSS 120B on Groq.
- Deployed on Streamlit Cloud with per-session, in-memory vector stores so uploads are never persisted or shared; built a RAGAS harness (20 hand-written Q/A pairs, separate judge LLM) where the dense baseline scored 0.95 faithfulness.

sid.agent: Portfolio AI Agent | *Python, FastAPI, Groq, SSE, Vercel*

[GitHub](#) · [Live](#)

- Built and deployed a public portfolio agent that answers recruiter questions in real time, served as a FastAPI serverless function on Vercel with SSE token streaming from GPT-OSS 120B on Groq and a warm-up endpoint for cold starts.
- Wrote lightweight retrieval over a markdown knowledge base: core sections always load and the top 3 keyword-scored topic sections are added per query, keeping prompts under Groq's token limit; guarded by per-IP rate limiting and input validation.

ACHIEVEMENTS & LEADERSHIP

- **Hackathons:** Anveshan Hackathon, IIT Guwahati (national level, 48 hours), built an ML model for fake job-posting scam detection; also competed in the Adobe Hackathon and 2x HackIIITV, shipping working prototypes in 24 to 36 hours.
- **Leadership:** Paid Teaching Assistant & Football Mentor, IIIT Vadodara (Aug 2025 – Present); Photographer, Obscura Club.
- **Football:** 3x state-level player (SGFI National trials); 3rd, Inter-IIIT Sports Meet; 2nd, Concours; 2x runner-up, Ventura.